MATH 564 Wednesday Sept 10

Valve design Project

(18)	2.75	1.15	1.06	70.4
(19)	2.36	1.10	1.00	56.5
(20)	4.97	1.11	3.95	0.9

How many points are needed to uniquely determine a quadratic model?

$$f(x+p) = f(x) + p^T \underbrace{\nabla f(x)}_{3 \text{ unk.}} + \frac{1}{2} p^T \underbrace{\nabla^2 f(x)}_{6 \text{ unk.}} p$$

At what point would it make sense to employ a locally quadratic model? How can you "know" if such a model is likely to be useful?

BFGS Algorithm

Given: $x_0 \in \mathbb{R}^n$, $f: \mathbb{R}^n \to \mathbb{R}$, $\varepsilon > 0$

Set: $H_0 \leftarrow I_n$ *
$k \leftarrow 0$

Step 1: If $\|\nabla f_k\| < \varepsilon$ stop (or other criteria)

Step 2: Compute search direction $p_k = -H_k \nabla f_k$ Newton step for approx H.

Step 3: Solve $\alpha_k = \underset{\beta}{\operatorname{argmin}}\, f(x_k + \beta p_k)$ require SW

Step 4: Updates:

$$s_k = x_{k+1} - x_k$$
$$y_k = \nabla f_{k+1} - \nabla f_k$$
$$\rho_k = y_k^T s_k$$
$$H_{k+1} = \underbrace{(I - \rho_k s_k y_k^T) H_k (I - \rho_k y_k s_k^T)}_{\text{rank } 1} + \underbrace{\rho_k s_k s_k^T}_{\text{rank } 1}$$
$$k \leftarrow k+1$$

Step 5: Go to step 1

* Any pos def H_0 is ok, other good choices:

(a) $H_0 = |f(x_0)| I$ include some scale information
Best when $f(x) \geq 0$ & $f(x^*) \sim 0$

(b) $H_0 = |\text{typ} f| I$ user supplied scale information.

Performance Example

$$f(x) = \sum_{k=1}^{9} 10(x_{k+1} - x_k^2)^2 + (1 - x_k)^2, \quad x \in \mathbb{R}^{10}$$

$$x_{init} = (1/10, 2/10, 3/10, \ldots, 1) \qquad \alpha_0 = 1$$

	GD (Armijo)	GD (SW)	CG (PR)	BFGS (SW)
# iter	620	665	78	26
# f	4534	5563	648	57
# g	620	1368	194	51
# eval	10,734	19,243	2588	567
$\|x_k - x^*\|$	6.1E-3	3.2E-3	5.5E-5	8.4E-5
$\|g_k\|$	8.8E-3	4.5E-3	6.0E-4	1.1E-6

Some BFGS details.

we must have symmetric pos. def. H_k.

To do this H_{k-1} must be pos. def. and $\underbrace{y_k^T s_k > 0}_{\text{SW cond.}}$

If $y_k^T s_k$ is small (or even negative) then the Newton step becomes unreliable. So, the strategy is to find a "nearby" pos. def. matrix (a more stongly pos. def. H)

Suppose H has eigenvalues $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ with corresponding eigenvectors $u_1, u_2, \ldots u_n$.

If $\lambda_1 < 0$ then H is indefinite

If $|\lambda_1| \ll |\lambda_n|$ then H is poorly conditioned

cond # $c = \dfrac{|\lambda_n|}{|\lambda_1|}$.